

JACKSON SICHANGWA

Software Development | Web & Database Development | AI & IoT Solutions

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PROFESSIONAL SUMMARY

Highly motivated and analytically driven final-year Computer Science student at Africa University (CGPA: 3.48/4.00), with a strong academic foundation and hands-on practical experience in software development, web application development, database management, IoT systems and AI solutions. Demonstrated ability to design and deliver innovative, real-world technology solutions through academic projects including a Procurement Management System, an IoT-based Smart Greenhouse and an AI-powered Livestock Chatbot. Proficient in C#, Python, JavaScript, Node.js, SQL and MongoDB, with strong problem-solving, analytical and collaborative skills. A disciplined, creative and results-oriented professional committed to continuous learning and leveraging technology to drive meaningful impact.

CORE COMPETENCIES & SKILLS

Software & Web Development

- Software Development & Systems Design
- Web Application Development (Front-End & Back-End)
- Object-Oriented Programming (OOP)
- RESTful API Development & Integration
- Database Design & Management
- UI/UX Design & Prototyping (Figma)

Programming Languages & Frameworks

- C# (.NET)
- Python
- JavaScript (ES6+)
- HTML5 & CSS3
- Node.js & Express.js
- TensorFlow.js
- Bootstrap
- SQL & MongoDB

AI, IoT & Emerging Technologies

- Artificial Intelligence & Machine Learning (TensorFlow.js)
- IoT Systems Design & Automation (Arduino)
- Chatbot Development & Conversational AI
- Sensor Integration & Real-Time Monitoring

Tools, Platforms & Other Skills

- Git & GitHub (Version Control)
- Postman (API Testing)
- Visual Studio Code
- Canva & Adobe Photoshop
- Microsoft Office Suite (Word, Excel, PowerPoint)
- Graphic Design & Social Media Content
- Research Methods & Report Writing
- Data Collection, Monitoring & Evaluation

Professional Skills

- Analytical & Critical Thinking
- Problem-Solving & Innovation
- Attention to Detail & Accuracy
- Teamwork & Cross-functional Collaboration
- Communication & Client Interaction
- Adaptability & Continuous Learning

PROJECTS & EXPERIENCE

Africa University | Mutare, Zimbabwe

Capstone Project – Software Engineering 2 & Web Application Development

January 2026 – June 2026

- **Procurement Management System:** Designed and developed a full supplier management module using C# and MySQL, implementing core procurement workflows including supplier registration, purchase order management and inventory tracking.

- **Web Application Development:** Built and deployed web-based applications using Node.js, Express.js, JavaScript, HTML5 and CSS3, applying software engineering principles including systems analysis, design patterns and agile development methodologies.

Africa University | Mutare, Zimbabwe

IoT Project – Automated Greenhouse Monitoring & Control System (Team Member) *January 2026 – April 2026*

- **IoT System Design & Development:** Contributed to the design and build of a smart IoT-based greenhouse system using Arduino microcontrollers, environmental sensors, water pumps and fans for real-time climate monitoring and automated irrigation control.
- **Automation & Resource Efficiency:** Implemented automated decision logic to trigger irrigation and climate control based on sensor thresholds, reducing manual intervention and improving resource efficiency across greenhouse operations.

Personal Project | Zambia

AI Project – Livestock Farming Chatbot *2025 – Present*

- **Conversational AI Development:** Designed and developed an AI-powered chatbot providing 24/7 reliable information on animal health, breeding and disease management to support local livestock farmers — applying NLP concepts and TensorFlow.js for intelligent response generation.

Africa University | Mutare, Zimbabwe

Portfolio Website – Front-End Development *August 2025 – November 2025*

- Designed and developed a personal portfolio website using HTML5, CSS3 and JavaScript, showcasing academic projects, technical skills and professional achievements.

Family Legacy Zambia | Lusaka, Zambia

Monitoring & Evaluation Volunteer *January 2023 – June 2023*

- **Programme Monitoring & Data Management:** Monitored and evaluated the academic performance of 200+ students, maintaining accurate records, tracking progress and preparing structured reports to support programme decision-making and improvement.
- **Student Coordination & Stakeholder Engagement:** Coordinated student placements into organizational and government schools, managing enrolment follow-up and liaising with schools, parents and organisational staff to ensure successful transitions.
- **Data Quality & Reporting:** Supported programme monitoring activities by maintaining data quality standards and contributing structured analytical input that improved educational outcome reporting.

EDUCATION

Bachelor of Science Honours in Computer Science *2023 – Present*

Africa University, College of Engineering & Applied Sciences | Mutare, Zimbabwe

Cumulative GPA: 3.48 / 4.00 | Final Year Student

Relevant Coursework:

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|---------------------------------|---|
| • Software Engineering 1 & 2 | • Data Communication & Computer Networks |
| • Web Application Development | • Operating Systems & Computer Architecture |
| • Database Systems & Algorithms | • Systems Analysis Design & Implementation |
| • Artificial Intelligence | • Simulation & Modelling |

Graphic Design & Social Media Networking Certificate *February – August 2022*

ZAMCOM | Zambia

A-Level School Certificate *2019 – 2021*

Chilenje South Secondary School | Lusaka, Zambia

Subjects: Mathematics, Physics, Biology, Accounts, Chemistry, Commerce

CERTIFICATIONS & ACHIEVEMENTS

- Cybersecurity Essentials Certificate – 2025
- Leadership Certificate – 2025
- Graphic Design & Social Media Networking Certificate – ZAMCOM, 2022

REFERENCES

Professional references are available upon request. The following individuals have agreed to provide references:

Name	Title	Contact	Email
Dr. Zengeni Tendai	HOD – Dept. of AI, Software Engineering & Computer Science	Africa University	zengenit@africau.edu
Ms. Benga	Campus Life Coordinator	Africa University	clc@africau.edu
Mrs. Chungu	M&E Officer, Family Legacy Zambia	+260 967 434 968	chilufya.mwala@familylegacyzambia.org

All referees have been contacted and have consented to provide professional references upon request.